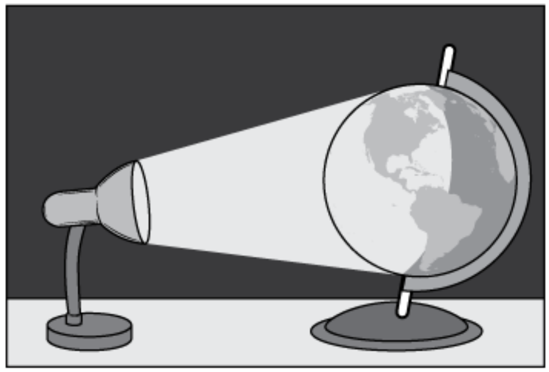
Tennessee Comprehensive Assessment Program

TCAP

Science
Grade 4
Practice Items

TN Department of Education

Students use a desk lamp to model light from the sun. The students use a globe to model Earth. The students shine the light from the desk lamp onto the globe.



The students want to model the reason that the sun appears to rise and set on Earth.

Which of these should the students do with the parts of the model?

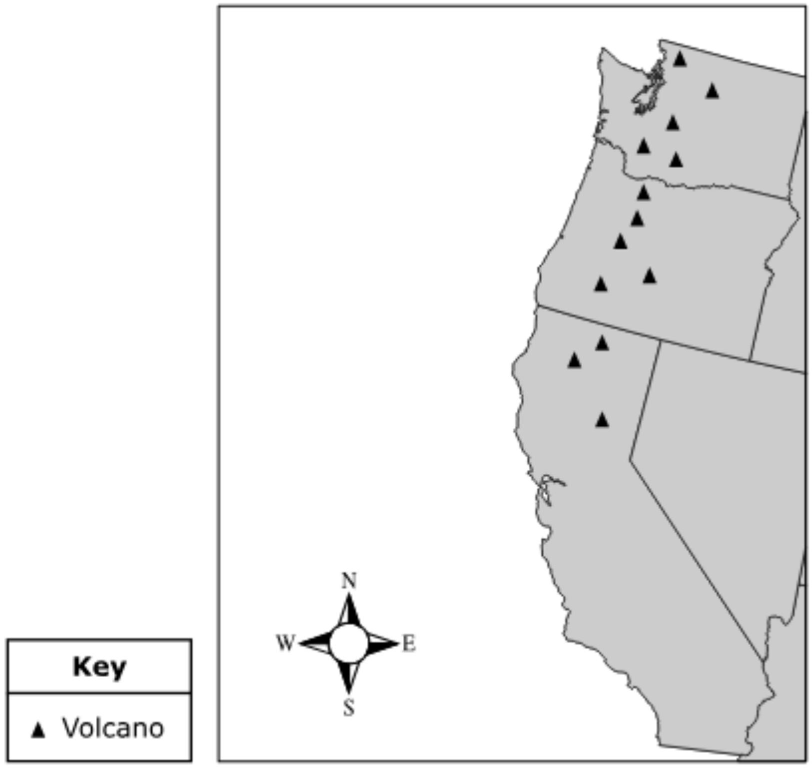
A. Spin the globe on its stand.

B. Turn the desk lamp on and off.

C. Lift the desk lamp above the globe.

D. Carry the globe in a circle around the desk lamp.

The map shows the location of some volcanoes in the United States.



What pattern is shown by the location of these volcanoes?

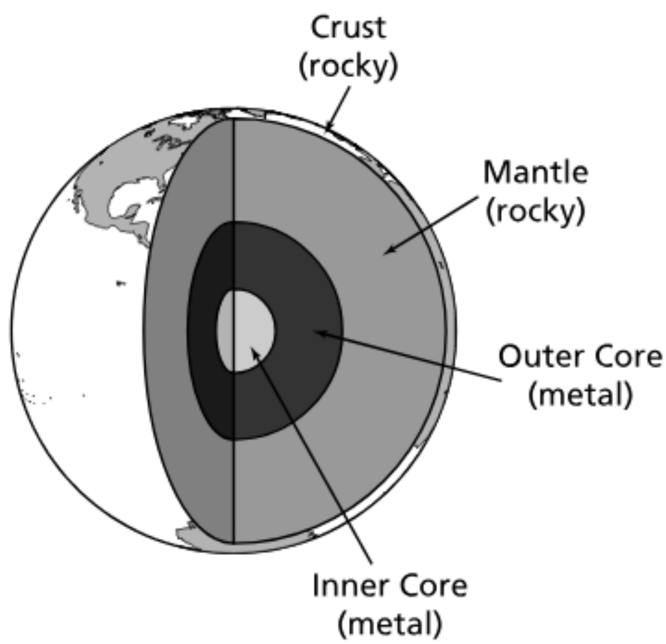
M. Volcanoes are usually located far from the ocean.

P. These volcanoes are centrally located around one city.

R. The source of these volcanoes is along the West Coast.

S. Volcanoes become inactive if they are under ocean water.

Earth has a magnetic field. Scientists think that this is partly because of magnetic materials found in some layers of Earth. The diagram shows the makeup of Earth's layers.



Which layers of Earth are **most likely** the reason for Earth's magnetic field?

A. Outer Core and Inner Core

B. Outer Core and Mantle

C. Crust and Inner Core

D. Crust and Mantle

Farmers often use fertilizer on their farms. Fertilizer is also used in gardens. Some people argue that fertilizer is bad for the environment.

Which effect of using fertilizer **best** supports the argument that fertilizer is bad for the environment?

M. Fertilizer stops soil erosion in fields.

P. Fertilizer puts nutrients back into the soil.

R. Fertilizer makes plants grow bigger and faster.

S. Fertilizer disrupts the healthy balance of the soil.

Students placed an equal-sized piece of a water plant into four test tubes filled with water. Each test tube was sealed with a stopper to keep the air inside. Students placed each test tube in light for different numbers of hours. The students recorded the number of oxygen bubbles in each test tube after they were taken out of the light. The students' data table is shown.

Plant Observations

Test Tube	Hours of Light	Amount of Oxygen Bubbles Present After 24 Hours
1	24	Largest ↓↓ Smallest
2	6	
3	3	
4	0	

Which **two** statements are **best** supported by the data?

A. Oxygen is made during photosynthesis.

B. Water is made during photosynthesis.

C. The smallest amount of oxygen is produced in light.

D. Light is needed for plants to complete photosynthesis.

E. Plants complete photosynthesis at night.

The diagram shows a food chain.

Food Chain

sunlight → small water plants → small water animals → fish → seals

Which **three** statements **best** describe how energy moves through this food chain?

M. Seals get energy from fish.

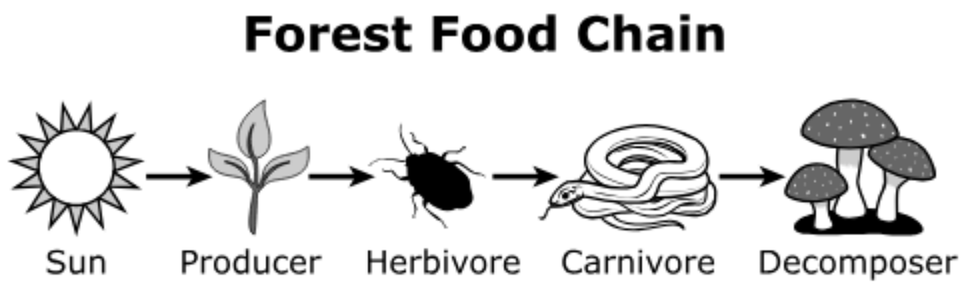
P. Fish get energy from seals.

R. Fish get energy from small water animals.

S. Small water plants get energy from sunlight.

T. Small water plants get energy from small water animals

A forest food chain is shown.



Which **two** statements **best** describe the movement of energy in this forest food chain?

A. The decomposer provides energy to the producer.

B. The carnivore provides energy to the producer.

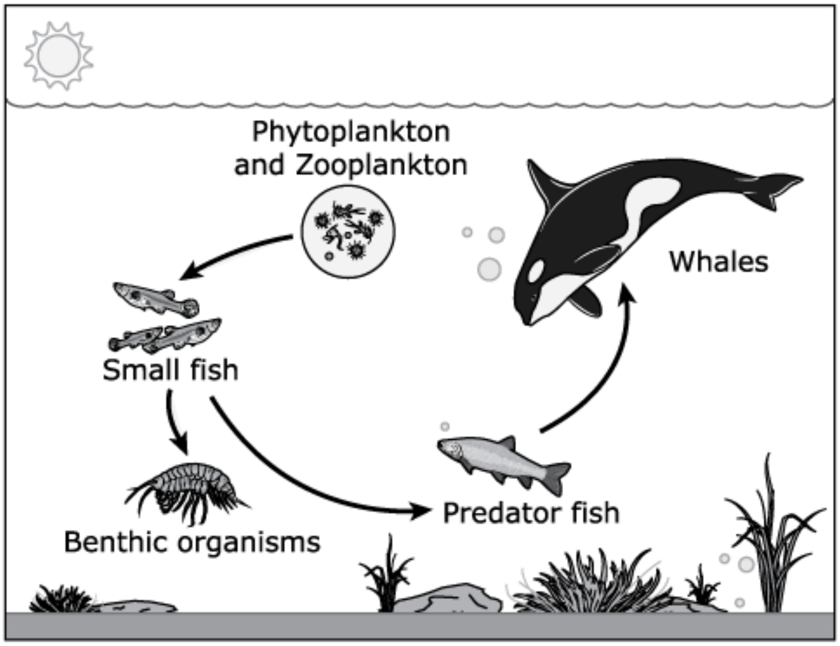
C. The carnivore provides energy to the herbivore.

D. The herbivore provides energy to the carnivore.

E. The producer provides energy to the herbivore.

A simple marine food web is shown.

Simple Marine Food Web



The small fish are important in this ecosystem because they

M. give energy directly to plankton and benthic organisms.

P. get energy from predator fish and benthic organisms.

R. give energy to zooplankton and get energy from predator fish.

S. get energy from both types of plankton and give energy to predator fish.

Kudzu is a vine that grows in Tennessee. Kudzu is not native to Tennessee. The vine has caused problems for the ecosystem. Scientists want to decrease the kudzu population. They also want to improve the health of the ecosystem. Which of these should scientists do to get rid of kudzu and improve the health of the ecosystem?

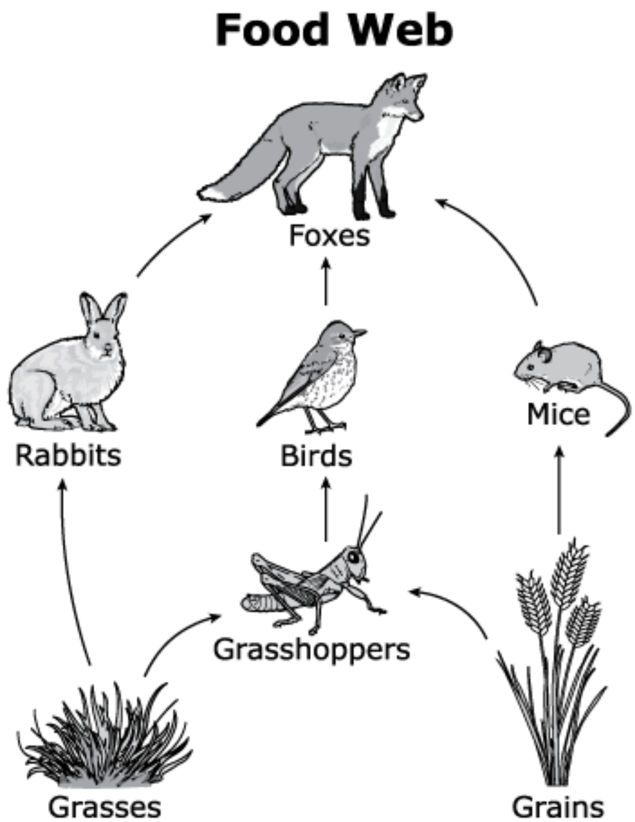
A. Pour concrete on the land that has kudzu.

B. Dig up all the land covered with kudzu.

C. Bring animals that eat kudzu to the land where it grows.

D. Spray poison on the land where kudzu grows.

The diagram shows a food web. A food web shows feeding relationships among different organisms that live in the same habitat.



Suddenly, there were fewer grasshoppers. Based on this food web, which population of organisms will also decrease because of this change?

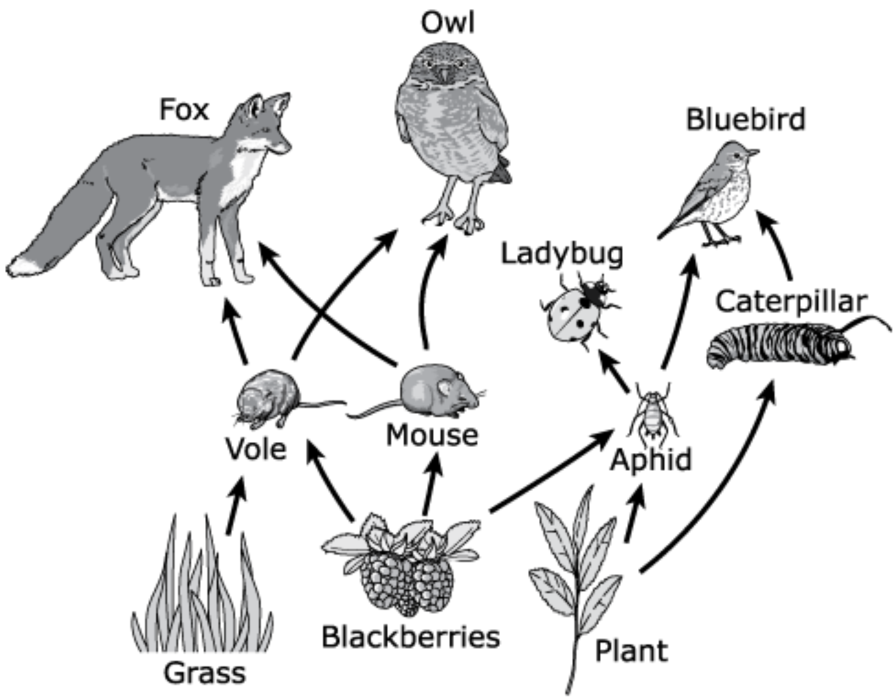
M. Grasses

P. Birds

R. Mice

S. Rabbits

European starling birds are a species that was brought to the United States in the 1800s. They were used to help control insect populations. The starlings eat berries and insects such as caterpillars. A food web is shown.



Based on the food web, which **three** ways would the introduction of the starling **most likely** affect this food web?

A. The caterpillar population would decrease.

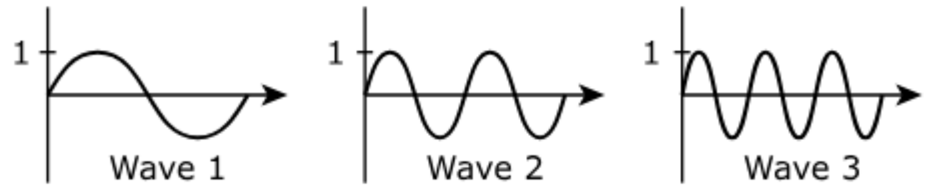
B. The mouse population would decrease.

C. The bluebird population would decrease.

D. The ladybug population would increase.

E. The fox population would increase.

Students compared three models of waves. The models are shown.



Which statement correctly compares two of the models?

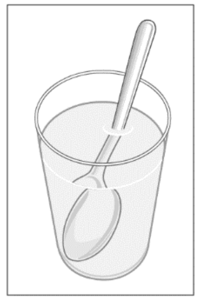
M. Wave 1 has a smaller amplitude than Wave 2.

P. Wave 2 has a bigger amplitude than Wave 3.

R. Wave 2 has a longer wavelength than Wave 1.

S. Wave 3 has a shorter wavelength than Wave 2.

A student looks at a spoon in a glass of water. The student observes that the spoon appears to be broken.



Why does the spoon look like it is broken?

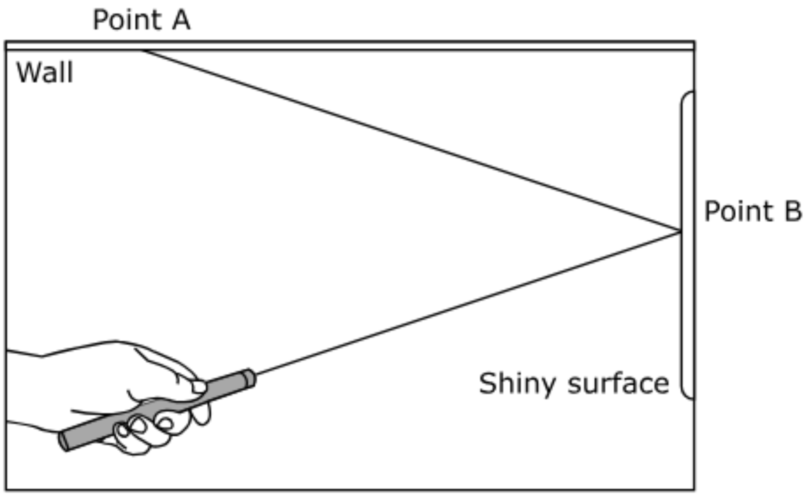
A. Light waves are bent by the water.

B. Light waves are enlarged by the water.

C. Light waves are absorbed by the water.

D. Light waves are reflected by the water.

A student uses a laser pointer and a shiny surface to model a reflection on the surface of water.



How does the model relate to the student seeing a reflection on the surface of the water?

M. The light being refracted at point B is like the refraction of her image through the water and into her eye.

P. The light being reflected at point B is like the light being reflected off the surface of the water and into her eye.

R. The light being transmitted at point A is like the transmission of her image through the water and into her eye.

S. The light being reflected at point A is like the light being reflected off the surface of the water and into her eye.

Metadata- Science Items

Page Number	Item Type	Key	TN Standards
1	MC	A	4.ESS1.2
2	MC	R	4.ESS2.2
3	MC	A	4.ESS2.4
4	MC	S	4.ESS3.2
5	MS	A,D	4.LS2.1
6	MS	M, R, S	4.LS2.2
7	MS	D,E	4.LS2.2
8	MC	S	4.LS2.3
9	MC	C	4.LS2.4
10	MC	P	4.LS2.4
11	MS	A,B,C	4.LS2.4
12	MC	S	4.PS4.1
13	MC	A	4.PS4.2
14	MC	P	4.PS4.2
Metadata Definitions			
Item Type	MC = Multiple Choice MS = Multiple Select		
Key	Correct answer		
TN Standards	Primary educational standard assessed.		